中熔电气
SINOFUSE

PV312-XL3-Q/T

产品规格书

PRODUCT SPECIFICATION

SINOFUSE

成为卓越的能量保护系统解决方案服务商
为整个人类的能量转换活动的安全运行提供保障
Provide excellent integrated solution of energy protection systems
Safeguard energy conversion processes for mankind

文件变更履历表/Revision Record

文件名称/File Name: PV312-XL3-Q/T
文件编号/File No.: ZR/SCG-0101

Version	Content	Execution Date	Author
S0	新制定/Newly developed	2025-7-16	Wang Chengyu
S1	更新参数表/Update parameter table	2025-8-7	Wang Chengyu

产品技术参数/Product Technical Parameters

- 参考标准/ Standards: GB13539, IEC60269
- 额定电压/Rated voltage: DC 1500V
- 额定电流/Rated current: 250A～630A
- 保护类别/Protection category: gPV
- 分断范围/Breaking range: 50kA(时间常数/Time constant：1～3ms)
- 认证/Accreditation: UL, TUV, CE
- 符合/Compliant: RoHS

No.	Part Number	Size	In (A)	$I^2t/(A^2s)$		Loss (w)	Resistance Value Mω (20℃)	Weight (g)	Max Package (pcs)	Remark
				Prearc	Clearing	In				
1	PV312-XL3-Q250A 1500V-N	3	250	110000	206000	50.6	0.6132±10%	2018±5%	6	M12 38±1N·m
2	PV312-XL3-Q315A 1500V-N		315	195000	366000	64.7	0.4599±10%			
3	PV312-XL3-Q350A 1500V-N		350	281000	527000	69	0.3833±10%			
4	PV312-XL3-Q400A 1500V-N		400	383000	718000	81.2	0.3285±10%			
5	PV312-XL3-Q450A 1500V-N		450	564000	1060000	88.7	0.2705±10%			
6	PV312-XL3-Q500A 1500V-N		500	781000	1460000	108	0.2190±10%			
7	PV312-XL3-Q550A 1500V-N		550	945000	1770000	121	0.2090±10%			
8	PV312-XL3-Q630A 1500V-N		630	1220000	2290000	143	0.1840±10%			
9	PV312-XL3-T250A 1500V-N		250	110000	206000	50.6	0.6132±10%	2020±5%		M10 21±1N·m
10	PV312-XL3-T315A 1500V-N		315	195000	366000	64.7	0.4599±10%			
11	PV312-XL3-T350A 1500V-N		350	281000	527000	69	0.3833±10%			
12	PV312-XL3-T400A 1500V-N		400	383000	718000	81.2	0.3285±10%			
13	PV312-XL3-T450A 1500V-N		450	564000	1060000	88.7	0.2705±10%			
14	PV312-XL3-T500A 1500V-N		500	781000	1460000	108	0.2190±10%			
15	PV312-XL3-T550A 1500V-N		550	945000	1770000	121	0.2090±10%			
16	PV312-XL3-T630A 1500V-N		630	1220000	2290000	143	0.1840±10%			

注/Note：

1. 功耗值测试个体为PV312-XL3-Q-N结构，以上数据表中的I^2t数据均基于标准试验条件下的I1(50kA)，I2 测试数据得出；

The power consumption value is tested for PV312-XL3-Q-N structure, the I^2t data in the above data sheet are based on the I1 (50kA), I2 test data under the standard test conditions.

2. 如需指示，型号后去掉-N，例: PV312-XL3-Q500A1500V(带指示件、基座结构)。

If you need to indicate, remove -N after the model number, Example: PV312-XL3-Q500A1500V (with indicator, pedestal structure).

产品特性/Product Features

本产品为光伏全范围保护用熔断器，具有低功耗、高分断能力、抗大幅电流冲击的特点。保护和隔离光伏汇流箱,以及保护其他适合的电路系统。

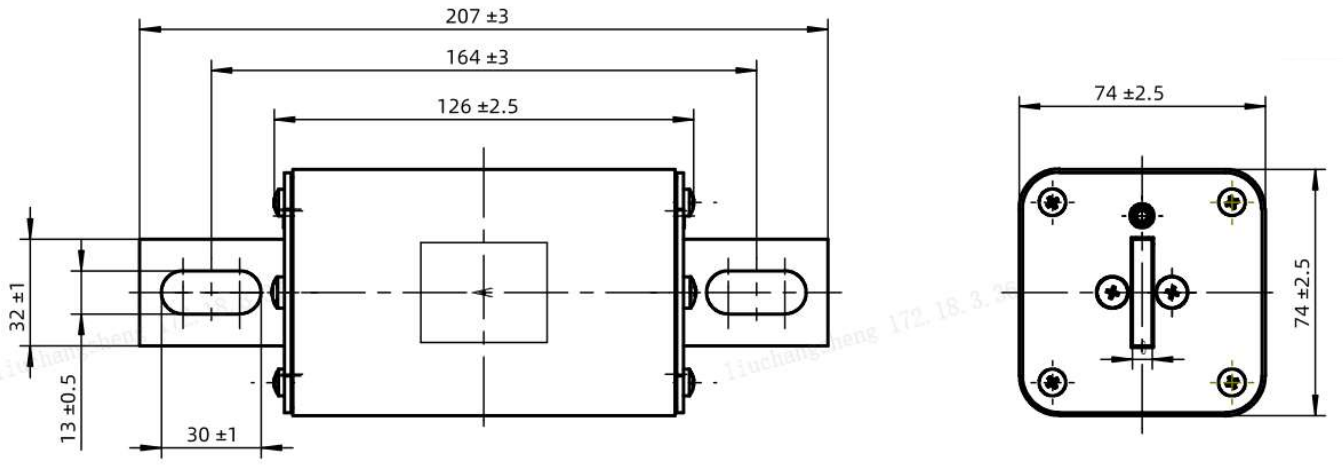
This product is a fuse for full range protection of photovoltaic, featuring low power consumption, high breaking capacity, and resistance to large current shocks. It protects and isolates the PV converter box, as well as protects other suitable circuit systems.

注：本规格书中所有试验数据均基于标准试验条件下的典型测试数据得出，试验参数和接线按照IEC 60269。

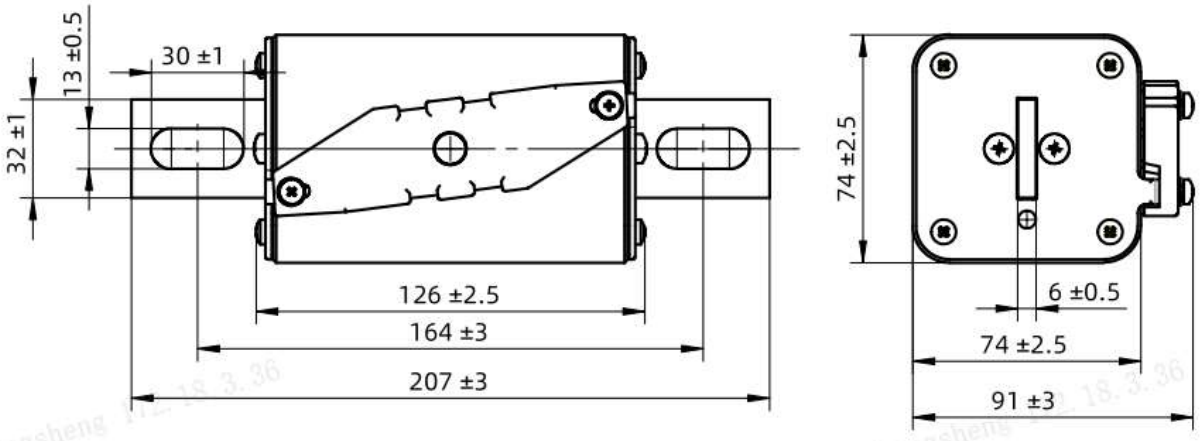
Note: All test data in this specification are based on typical test data under standard test conditions, and the test parameters and wiring are in accordance with IEC 60269.

产品外形尺寸/Dimensions(mm)

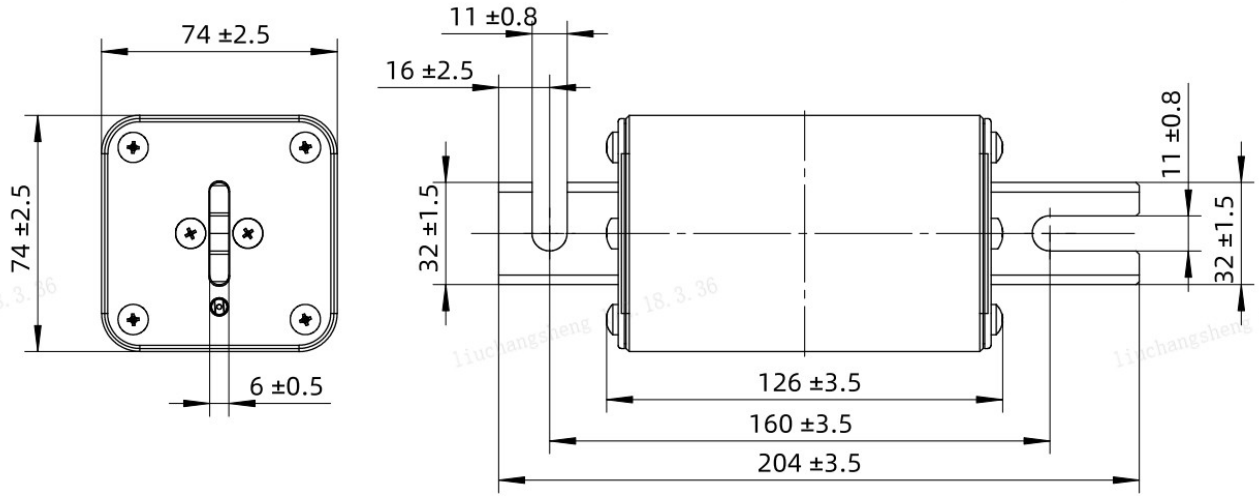
PV312-XL3-Q-N



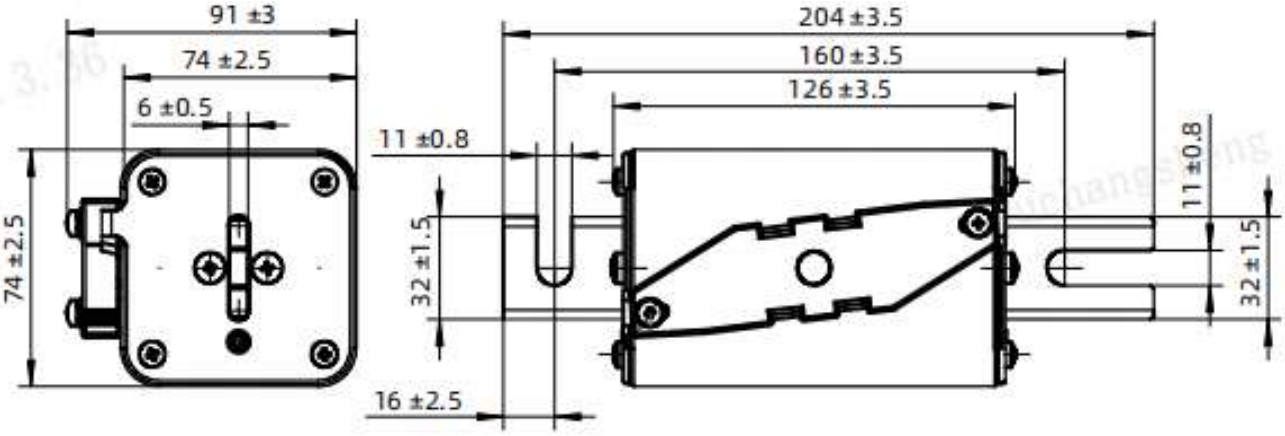
PV312-XL3-Q



PV312-XL3-T-N



PV312-XL3-T

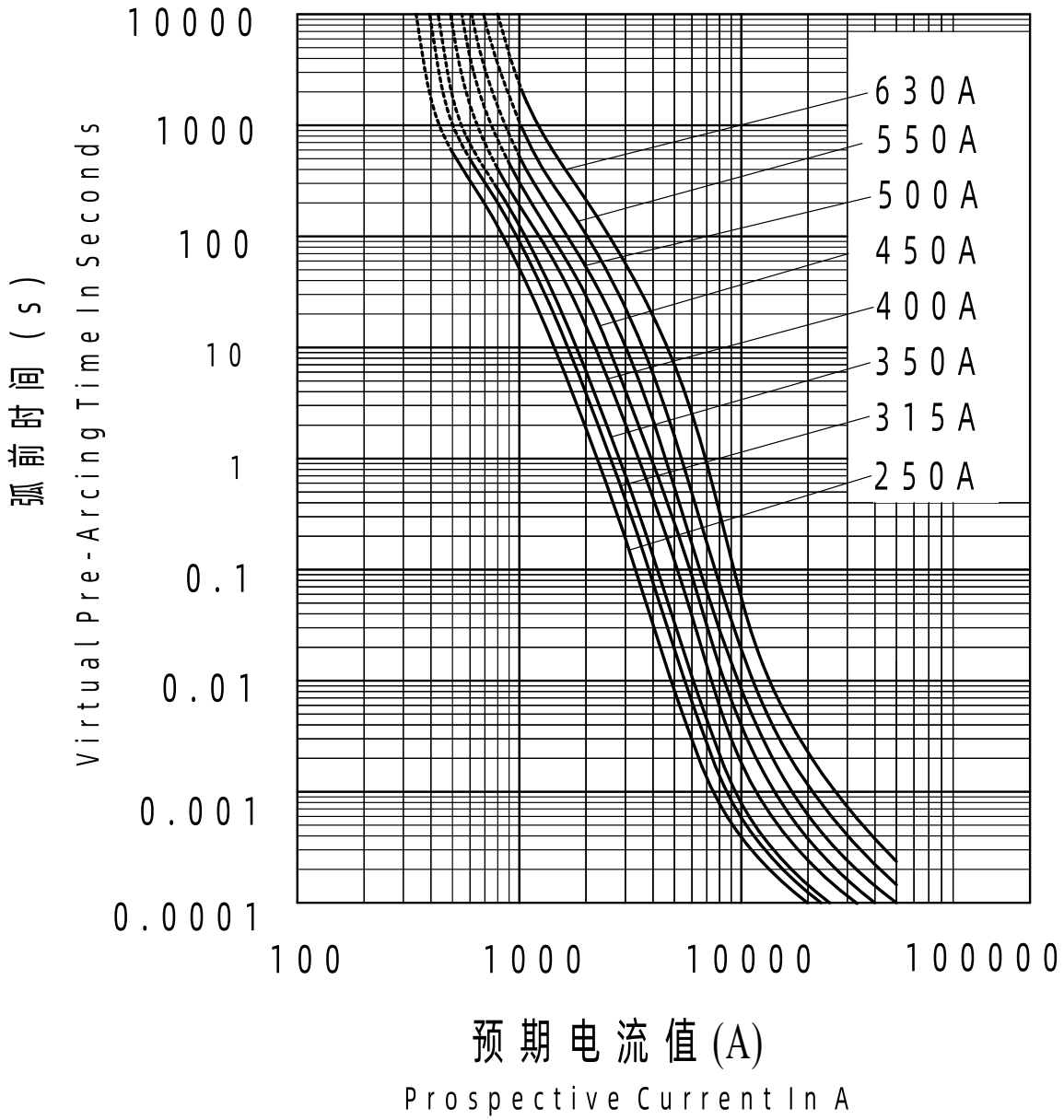


主要材质清单/Main Material List

No.	Part Name	Material	Electroplated
1	联结板/Coupling Plate	T2+H62	镀暗锡/Dark tin plated
2	瓷管/Porcelain Tube	A-75	/

产品特性曲线/Characteristic Curve

弧前I-t曲线/Pre-arcing I-t Curve



注/Note:

1. 特性曲线精度为电流方向误差±15%；/Characteristic curve accuracy is current direction error ±15%.
2. 最小分断电流：≥2In；/Minimum breaking current: ≥2In.
3. 曲线中100ms以下为等效弧前时间。/Below 100ms in the curve is the equivalent pre-arc time.

运输、存储/Transportation, Storage

运输/Transportation

避免过度挤压或冲击：在运输过程中，需要避免对熔断器施加过度挤压或直接冲击，这可能会损坏熔断器的外壳、接线端子或内部元件，从而影响其性能和安全性。运输过程中应避免雨雪侵袭和机械损伤。

Avoid excessive compression or shock: During transportation, it is necessary to avoid applying excessive compression or direct shock to the fuse. This may damage the fuse housing, terminals or internal components, thus affecting their performance and safety. Avoid rain, snow and mechanical damage during transportation.

防潮防湿：熔断器应远离潮湿和湿润的环境，以防止受潮和腐蚀。必要时，可以使用防潮包装材料或采取其他措施来保护熔断器免受潮湿影响。

Moisture and humidity protection: Fuses should be kept away from damp and wet environments to prevent moisture and corrosion. If necessary, moisture-proof packing materials may be used or other measures taken to protect the fuse from moisture.

存储/Storage

1.产品存储温度：-40℃～85℃；
Storage Temperature: -40℃～85℃.
2.包装存储温度：-40℃～70℃；
Package Temperature: -40℃～70℃.
3.包装存储湿度：相对湿度≤90%，无凝露。
Package Humidity: RH≤90%, no dewing.

使用条件/Use Conditions

熔断器所有试验均在标准测试条件下进行，因实际使用条件和标准测试条件存在差异，在选用时需考虑其影响。当实际使用条件满足允许使用条件时，需按照对应修正曲线对熔断器的额定值进行修正；当实际使用条件不满足允许使用条件时，请咨询我司。

All tests for the fuse are conducted under standard testing conditions, there is the difference between the actual use conditions and the test conditions, the influence of which should be taken into account when selecting the fuses. When actual usage condition meets tolerable range, the rated value should be corrected according to the corresponding correction curve. Otherwise, please consult Sinofuse.

大气条件/Atmosphere

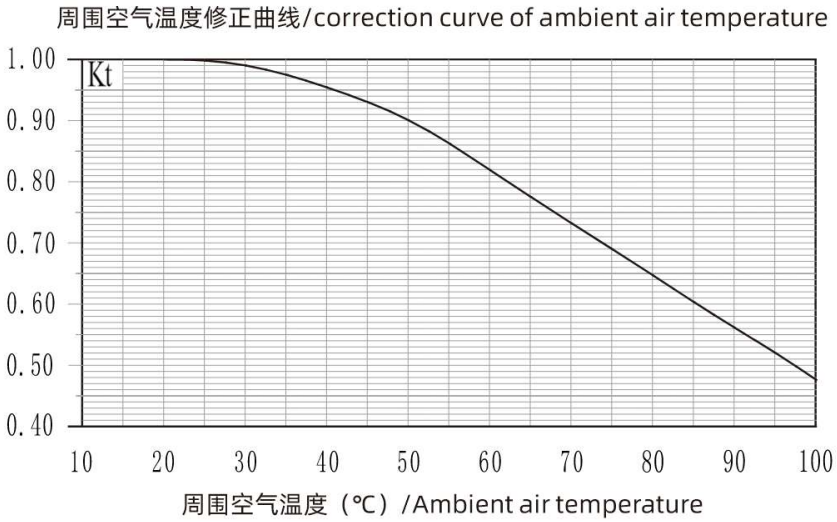
空气是干净的，相对湿度在最高温度为40℃时不超过50%。在较低的温度下可以有较高的相对湿度，例如，在20℃时，相对湿度可达90%。
For clean air, maximum 50% RH at 40℃. Higher RH is allowed when temperature is lower, e.g. maximum 90% at 20℃.

周围空气温度/Ambient Air Temperature

标准测试条件：20℃；
Standard Testing: 20℃.
允许使用条件：-40℃～100℃；
Tolerable Range: -40℃～100℃.

熔断器在周围空气温度20℃以上工作，额定电流按照下面曲线进行修正；温度修正时一般不考虑增加熔断器额定电流（除非20℃以上不是工作范围）。

When the fuse operates at an ambient air temperature above 20℃, the rated current shall be adjusted according to the following curve. temperature derating typically does not consider increasing the fuse's rated current (unless above 20℃ is not within the operating range) .



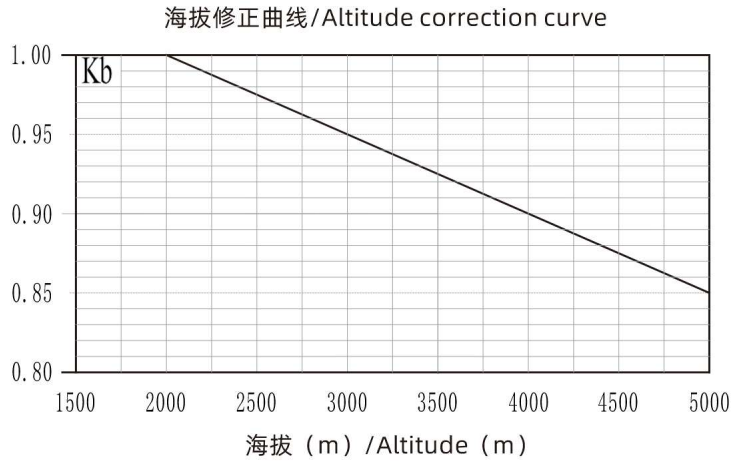
海拔/Altitude

标准测试条件：海拔≤2000m；

Standard Testing: Altitude≤2000m.

允许使用条件：海拔≤5000m，不同海拔下熔断器额定电流按照下面曲线进行修正。

Tolerable Range: Altitude≤5000m; rated current should be corrected according to the following curve at different altitudes.



风冷/Air Cooling

标准测试条件：无风，周围1m以内无发热器件；

Standard Testing: no ventilation, no heat source within 1m.

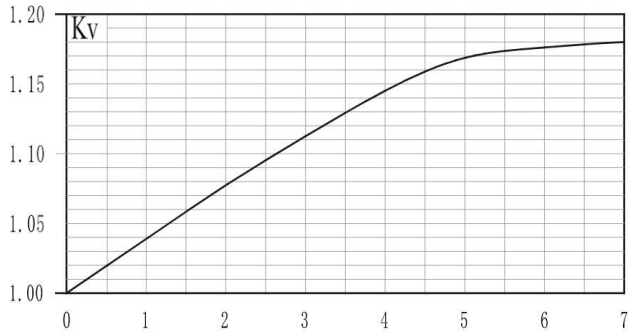
允许使用条件：允许有风及周围有发热器件，但需考虑其影响；

Tolerable Range: ventilation and heat source allowed, but their impacts should be considered.

强制风冷：采用强制风冷可以改善熔断器的散热，提升熔断器的额定电流，强制风冷条件下熔断器额定电流按照下面曲线进行修正。

Forced Air Cooling：Forced air cooling improves the heat dissipation of fuses, enabling an increase in their rated current. Under forced air cooling conditions. Rated current should be corrected according to the following curve.

强制冷却风速修正曲线/Correction curve of forced cooling air velocity



强制冷却风速(m/s)/Forced cooling air velocity(m/s)

铜导线/铜排尺寸/Copper wire/Busbar size

标准测试条件：铜导线/铜排尺寸满足标准要求；

Standard Testing: The size of the copper wire/busbar complies with the standard requirements.

允许使用条件：推荐铜导体截面不小于标准截面的50%（标准截面以实际通流值为前提），其他材质导体需换算为铜导体的尺寸进行校核。

Tolerable Range: It is recommended that the cross-sectional area of the copper conductor should not be less than 50% of the standard cross-sectional area (the standard cross-section is based on actual current-carrying capacity). For conductors made of other materials, their dimensions must be converted to equivalent copper conductor dimensions for verification.

标准导线选用表（参考GB/T13539.1）/Standard Conductor Selection Chart（GB/T13539.1）

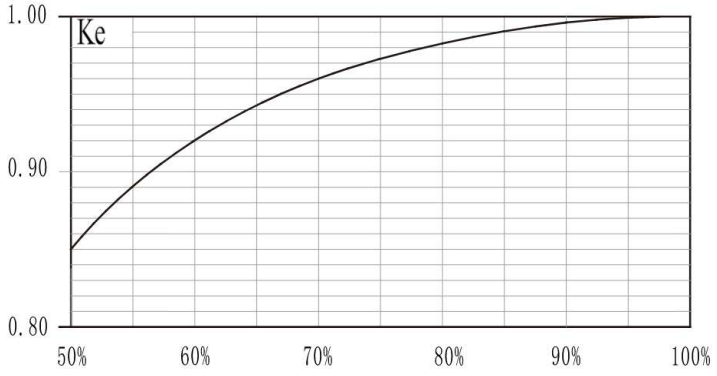
额定电流 A Rated Current	导线截面积 mm² Conductor Cross-sectional Area
2、4、6	1
8、10、12	1.5
16、20	2.5
25	4
32	6
40、50	10
63	16
80	25
100	35
125	50
160	70

额定电流 A Rated Current	导线截面积 mm² Conductor Cross-sectional Area
200、225	95
250、300	120
315、350	185
400、425	240
500	300
630	370
800	480
1000	600
1250	800
> 1250	（1.0~1.6）A/mm²

不同铜导体尺寸下熔断器的额定电流按照下面曲线进行修正

Rated current should be corrected according to the following curve

接线修正曲线/ Wiring correction curve



实际截面/标准截面/ Actual cross-section/standard cross-section

机械振动/机械冲击/Mechanical Vibration/Mechanical Shock

非底座安装类熔断器有良好的抗振动和冲击的耐受能力（满足GB/T 21563 轨道交通机车车辆设备 冲击和振动试验1类B级）。

Non base-mounted type fuse has great resistance to vibration and shock (meets Standard GB/T 21563 Shock and Vibration Test Ⅰ.B Level).

关于底座安装类安装熔断器的机械振动/冲击能力，以及方体瓷管类产品机械振动/机械冲击更为强烈的应用环境，请咨询我司。

For the mechanical vibration/shock capabilities of base-mounted fuses and harsher conditions for square porcelain tube products, please consult Sinofuse.

污染等级/Pollution Level

熔断器满足污染等级2（详见GB 14048.1-2023 7.1.3.2）。

Fuse meets pollution level Ⅱ (see GB 14048.1-2023 7.1.3.2 for details).

熔断器安装条件/Mounting Condition

使用规格书推荐扭矩安装，安装方式及注意事项见相应结构熔断器安装说明。

Use the torque specified in the specification. Refer to the fuse installation guide for methods and precautions.